

Interpretable Classification of Traditional Chinese Musical

Instruments Using Timbral Audio Features

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1 Project Overview

This project aims to automatically identify traditional Chinese musical instruments from short audio clips. While different instruments possess unique timbres, identifying them can be challenging for listeners unfamiliar with traditional Chinese music. A reliable and easily understandable identification system can support music information retrieval, music education, and the digital exploration of Chinese musical culture.

The project will first reproduce the published MFCC-KNN benchmark model for all eleven instruments in the ChMusic dataset using a five-second audio clip. Subsequently, the benchmark model will be further extended by combining MFCC features with interpretable spectral and temporal features. The original and improved methods will be compared using classification accuracy, macro-F1 score, and confusion matrix. In addition to predicting the instrument category, the proposed system will also provide a concise explanation of the measured timbre features. Detailed demonstrations will focus on the erhu, dizi, and guzheng, as they employ different sound production mechanisms and exhibit significant timbre differences.

2 Background and Motivation

Instrument recognition is an important application of music information retrieval, which can automatically organize, search and analyze the instruments contained in audio recordings. However, many widely used instrument datasets mainly focus on Western instruments, while there are relatively few public resources available for research on traditional Chinese music. To make up for this deficiency, Gong et al. developed the ChMusic dataset, which contains 55 recordings of 11 traditional Chinese instruments [1]. They also provided a benchmark method based on MFCC-KNN and reported a classification accuracy of 94.15% at K=15.

Recent studies have explored more advanced methods for addressing this problem.

Li and Zhang used Mel spectrograms and convolutional neural networks to identify traditional Chinese musical instruments [2]. Yang et al. further demonstrated that combining multiple time-frequency features can provide richer information for instrument classification [4]. However, high classification accuracy alone cannot explain which sound features have an impact on prediction. Therefore, Chen et al. investigated methods for interpreting instrument recognition models using multiple spectrogram representations and heatmaps [3]. These studies show that classification performance is just as important as interpretability when developing audio recognition systems.

I chose this topic because traditional Chinese musical instruments possess unique timbres that reflect different physical methods of sound production. Exploring these differences provides an interesting approach to understanding the representation of musical sounds in the time and frequency domains. It also allows me to apply concepts introduced in the ELEC5305 course, including waveforms, spectrograms, spectra, MFCCs, and frame-based audio features. By first reproducing existing methods and then investigating additional timbre features, I hope to gain a clearer understanding of how signal processing techniques represent and differentiate musical sounds. This project also provides an opportunity to showcase traditional Chinese musical instruments through measurable audio characteristics, rather than relying solely on subjective descriptions.

3 Proposed Methodology

3.1 Tools and Platforms

MATLAB will serve as the primary platform for audio preprocessing, feature extraction, classification, and result visualization. The Audio Toolbox, Signal Processing Toolbox, and Statistics and Machine Learning Toolbox will be used for audio feature extraction and the implementation of the K-Nearest Neighbor (KNN) classifier. Project code, documentation, and experimental results will be maintained in a public GitHub repository.

3.2 Signal Processing Techniques

This project will first reproduce the MFCC-KNN method based on the ChMusic dataset [1]. Each audio recording will be converted to mono, resampled to a consistent sampling rate, and segmented into non-overlapping 5-second segments. Twenty Mel-frequency cepstral coefficients (MFCCs) will be extracted from the short frames. KNN will classify each frame, determining the final instrument label for each audio segment through majority voting.

Subsequently, this reproduction method will be further extended by combining MFCCs with multiple interpretable audio features, including spectral centroid, spectral bandwidth, spectral roll-off, spectral contrast, zero-crossing rate, and root-mean-square (RMS) energy. These features describe characteristics such as

brightness, frequency distribution, noise level, and signal energy. The original method and the extended method will be compared by classification accuracy, macro F1 score, and confusion matrix. Meanwhile, spectrograms and feature distributions will also be used to interpret timbre differences between selected instruments.

3.3 Data Source

This project will use the publicly available ChMusic dataset, which contains 55 recordings of 11 traditional Chinese musical instruments [1]. According to the original study, the first to fourth recordings of each instrument will be used for training, and the fifth recording will be used for testing. To prevent data leakage, all segments of the same original recording will be kept in the same dataset partition. Although the classification experiment will include all 11 instruments, the erhu, dizi, and guzheng will be selected for more detailed timbre analysis.

4 Expected Outcomes

The expected outcome is a working MATLAB prototype system capable of identifying eleven traditional Chinese musical instruments from a five-second audio clip and providing a brief analysis of their timbre characteristics. Baseline and extended methods will be evaluated using classification accuracy, macro-F1 scores, and confusion matrices, supplemented by selected spectrograms and feature plots. A public GitHub repository will contain the MATLAB code, project report, README document, and example results. A short demo video will demonstrate the complete workflow from loading the audio clip and predicting the instruments to displaying their audio analysis.

5 Project Timeline

Week	Tasks
Weeks 1–4	Review the literature, determine the project scope, complete the proposal and create the GitHub repository.
Weeks 5–6	Prepare the ChMusic dataset and develop the audio preprocessing workflow in MATLAB.
Weeks 7–8	Reproduce the published MFCC-KNN baseline and evaluate its classification performance.
Weeks 9–10	Add the interpretable timbral features and compare the extended method with the baseline.
Weeks 11–12	Analyse the results, prepare timbre examples for selected instruments and complete the MATLAB prototype and GitHub documentation.
Week 13	Finalise the report, record the demonstration video and complete the final submission.

6 References

- [1] X. Gong, Y. Zhu, H. Zhu, and H. Wei, "ChMusic: A traditional Chinese music dataset for evaluation of instrument recognition," in Proceedings of the 4th International Conference on Big Data Technologies (ICBDT ' 21), New York, NY, USA: Association for Computing Machinery, 2022, pp. 184 – 189, doi: 10.1145/3490322.3490351.
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